

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:02-01-2025**

Time: 3HRS **GTM-9** **Max. Marks: 300**

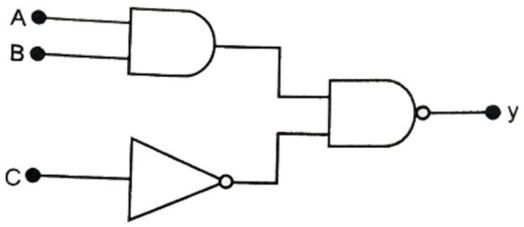
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

01. In the following circuit the output y becomes zero for the inputs



A) $A=1;B=0;C=0$

B) $A=1;B=1;C=0$

C) $A=0;B=1;C=1$

D) $A=0;B=0;C=0$

APEX JEE-MAIN 2025

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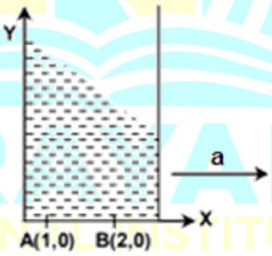
02. Position of particle is given by $x = t^3 - 4t^2 + 5t + 9$. What would be the distance travelled by particle from instant $t = 0$ to instant when particle changes its direction of velocity for last time.

A) 1.85 m B) 2.15 m C) 2m D) 2.85 m

03. A silver ball is suspended by a string in a vacuum chamber and ultraviolet light of wavelength 200 nm is directed at it. What electrical potential will the ball acquire as a result of it, if work function of silver is 4.7 eV $\left[h = 6.6 \times 10^{-34}\text{ J-s and } c = 3 \times 10^8\text{ m/s}\right]$

A) $1.5V$ B) $3.5V$ C) $4V$ D) $0.5V$

04. Fluid of density 10^3 kg/m^3 is kept in container. Container is moving with acceleration of 10 ms^{-2}. Find the magnitude of pressure (in Pa) difference between A and B as shown in figure. Distances are in meters. (Take $g = 10\ m/s^2$)



A) 10^4 B) 0 C) $\sqrt{2} \times 10^4$ D) 2×10^4

05. The wavelength of radiation emitted is λ_0 when an electron jumps from the third to the second orbit of hydrogen atom. If the electron jumps from the fourth to the second orbit of the hydrogen atom, the wavelength of radiation emitted will be

A) $\frac{16}{25}\lambda_0$ B) $\frac{20}{27}\lambda_0$ C) $\frac{27}{20}\lambda_0$ D) $\frac{25}{16}\lambda_0$

06. A time varying force is applied on a block placed over a rough surface as shown in figure. Let θ be the angle between net contact force on the block and the normal reaction. Then with time, θ will



Rough

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A) remain constant

B) first increase to a maximum value (say θ_{max}) and then becomes constant in a value slightly less then θ_{max}

C) first decrease to a minimum value (say θ_{min}) and then becomes constant in value slightly more then θ_{min}

D) None of the above

07. A body is projected vertically upwards from the surface of the earth with a velocity equal to half of escape velocity of the earth if R is radius of the earth, maximum height attained by the body from the surface of the earth is

A) $h=\frac{R}{4}$ B) $h=\frac{R}{2}$ C) $h=\frac{R}{3}$ D) $h=R$

08. Find the magnetic moment of a thin round loop with current if the radius of the loop is equal to $R=100\,mm$ and the magnetic induction at its centre is equal to $B=6\mu T$.

A) $3\times10^{-2}\,Am^2$ B) $4\times10^{-3}\,Am^2$ C) $5\times10^{-2}\,Am^2$ D) $2\times10^{-2}\,Am^2$

09. The displacement of two identical particles executing SHM are represented by equations $x_1=4\sin\left(10t+\frac{\pi}{6}\right)$ and $x_2=5\cos\omega t$

For what value of ω , the energy of both the particles become equal?

A) 16 unit B) 6 unit C) 4 unit D) 8 unit

10. An ideal gas mixture filled inside a balloon expands according to the relation $PV^{\frac{2}{3}}=$Constant. The temperature inside the balloon is

A) Increasing B) Decreasing C) Constant D) Can't be said

11. A plane electromagnetic wave of frequency $200\,MHz$ is travelling in vacuum along the positive x-direction. At a particular point in space and time magnetic field is $B=20\times10^{-8}\hat{k}\,T$. What is the electric field at that point (speed of light $c=3\times10^8\,ms^{-1}$)

A) $60.0\hat{k}\,V/m$ B) $6\hat{k}\,V/m$ C) $60\hat{j}\,V/m$ D) $6\hat{j}\,V/m$

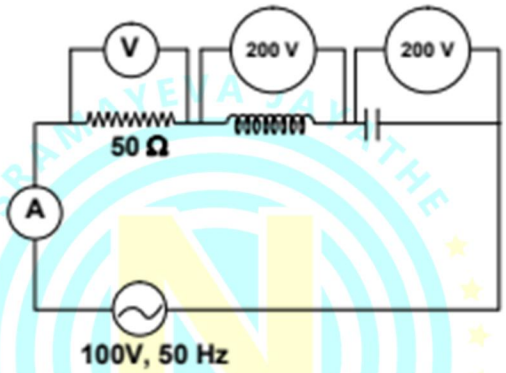
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12. The dimensional formula for the physical quantity $\frac{EB}{\mu}$ is (here, E-Intensity of electric field, B-magnetic induction and μ-absolute permeability).

A) ML^0T^{-3} B) ML^2T^{-2} C) $M^2L^3T^{-3}$ D) ML^2T^{-3}

13. In series LCR circuit voltmeter and ammeter reading are



A) V = 250 V, I = 4A B) V = 150 V, I = 2A

C) V = 1000V, I = 5A D) V = 100 V, I = 2A

14. The rms speed of oxygen molecule in a vessel at particular temperature is $\left(1+\frac{1}{x}\right)^{\frac{1}{2}}V$, where 'V' is the average speed of the molecule at the same temperature. The value of 'x' will be $\left(\pi=\frac{22}{7}\right)$.

A) 2.8 B) 3.8 C) 5.6 D) 6.6

15. Angle of minimum deviation is equal to the angle of prism A of an equilateral glass prism. The angle of incidence at which minimum deviation will be obtained is

A) 60° B) 30° C) 45° D) $\sin^{-1}(2/3)$

16. 1 cm on the main scale of a vernier calipers is divided into 10 equal parts. If 10 divisions of vernier coincide with 8 small divisions of main scale, then the least count of the vernier calipers will be:

A) 0.01 cm B) 0.02 cm C) 0.05 cm D) 0.005 cm

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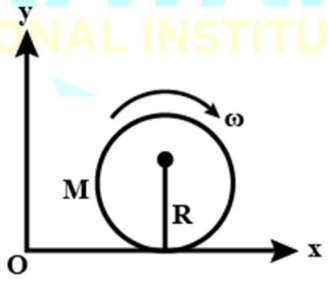
17. The period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by.

A) $2\pi\sqrt{\frac{L}{g\cos\alpha}}$ B) $2\pi\sqrt{\frac{L}{g\sin\alpha}}$ C) $2\pi\sqrt{\frac{L}{g}}$ D) $2\pi\sqrt{\frac{L}{g\tan\alpha}}$

18. Positive and negative point charges of equal magnitude are kept at $\left(0,0,\frac{a}{2}\right)$ *and* $\left(0,0,\frac{-a}{2}\right)$ respectively. The work done by the electric field when another positive point charge is moved from $(-a,0,0)$ *to* $(0,a,0)$ is.

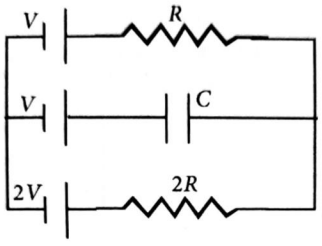
A) Positive
B) Negative
C) Zero
D) Depends on the path connecting the initial and final positions

19. A disc of mass M and radius R is rolling with angular speed ω on a horizontal *x-z* plane as shown in figure. The magnitude of angular momentum of the disc about the origin O is.



A) $(1/2)MR^2\omega$ B) $(3/2)MR^2\omega$ C) $MR^2\omega$ D) $2MR^2\omega$

20. In the given circuit with steady current the potential drop across the capacitor must be:



A) V B) $V/2$ C) $V/3$ D) $2V/3$

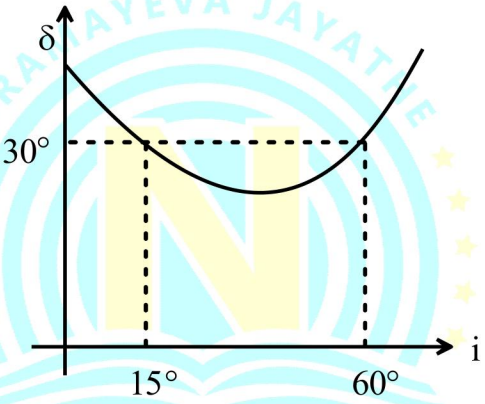
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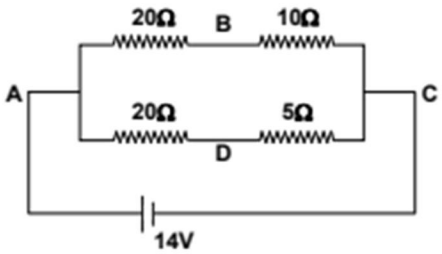
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Figure shows the graph of angle of deviations δ versus angle of incidence i for a light ray striking a prism. The prism angle is _____ (in degree).



22. The value of resistance $(\text{in } \Omega)$ should be connected in parallel with $20\,\Omega$ resistor (in ADC branch) so that potential difference between B & D may be zero.



23. Consider a body of mass $1.0\ kg$ at rest at the origin at time $t = 0$. A force $\vec{F} = \left(\alpha t\,\hat{i} + \beta\,\hat{j}\right)$ is applied on the body, where $\alpha = 1.0\text{N s}^{-1}$ and $\beta = 1.0\text{N}$. The torque acting on the body about the origin at time $t = 1.0\ s$ is $\vec{\tau}$. $\left|\vec{\tau}\right| = \frac{1}{x} Nm$ then find x

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24. A wire of length L metre carrying a current I ampere is bent in the form of a circle. The magnitude of its magnetic moment is $M = \frac{L^x i}{4\pi}$ then find x _____

25. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure $8\ N/m^2$. The radii of bubbles A and B are 2 cm and 4 cm respectively. Surface tension of the soap-water used to make bubbles is $0.04\ N/m$. The value of n_B / n_A is _____ (where n_A and n_B are the number of moles of air in bubbles A and B, respectively). [Neglect the effect of gravity.]

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Dry air was passed successively through a solution containing $5\,g$ of solute (non-volatile and non-electrolytic) in $90\ ml$ H_2O and then through pure H_2O. The loss in weight of solution was $2\,g$ and that of pure solvent $0.05\,g$. The mol. wt. of solute (in gm) is: (Given: density of water $= 1g/cc$)

A) 60 B) 40 C) 80 D) 20

27. For the fuel cell reaction $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$; $\Delta_f H^0_{298}(H_2O, l) = -285.5$ kJ/mol

The values of ΔS^0_{298} and ΔG^0_{298} for the given fuel cell reaction respectively are

Given : $O_{2(g)} + 4H^+_{(aq)} + 4e^- \rightarrow 2H_2O_{(l)}, E^0 = 1.23\ V$

A) -0.323 kJ/K, -474.78 kJ B) -0.635 kJ/K, -463.78 kJ

C) 3.51 kJ/K, -463.78 kJ D) -0.323 J/K, -474.78 kJ

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28. At $25°C$, Solubility of AgCl is S_1, S_2, S_3 and S_4 respectively in pure H_2O, 0.3M $BaCl_2$, 0.2M $AgNO_3$ and 0.5M HCl, then decreasing order of solubility is

(Given: $k_{sp}AgCl = 1\times10^{-10}$ at $25°C$)

A) $S_2 > S_1 > S_4 > S_3$ B) $S_1 > S_4 > S_3 > S_2$

C) $S_1 > S_3 > S_4 > S_2$ D) $S_3 > S_1 > S_4 > S_2$

29. $N_2O_{5(g)}$ decomposes to $NO_{2(g)}$ and $O_{2(g)}$ follows 1st order kinetics. After 50 minutes the pressure inside the vessel increases from 50mm Hg to 87.5 mm Hg. The pressure of the gaseous mixture after 100 minutes at constant temperature will be

A) 175.0 mm Hg B) 116.25mm Hg

C) 136.25 mm Hg D) 106.25 mm Hg

30. 1.0 molal aqueous solution of an electrolyte A_2B_3 is 70% ionized. The boiling point of the solution at 1 atm is $\left(K_{b(H_2O)} = 0.5\,k\,kg\,mol^{-1}\right)$ (BP of water is 373K at 1 atm)

A) 371.1 K B) 374.9 K C) 376.4 K D) 373.76 K

31. Find $\Delta_r H^0$ for the reaction $4HCl_{(g)} + O_{2(g)} \rightarrow 2Cl_{2(g)} + 2H_2O_{(g)}$ at 300 K. Assume all gases are ideal.

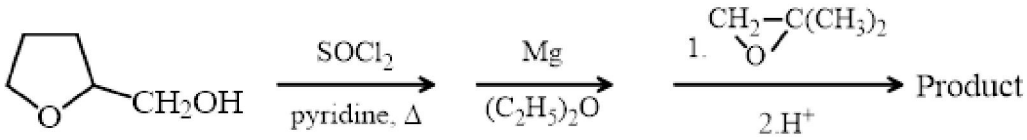
Given : $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$; $\Delta_r H^0_{300} = -184.5\,kJ$

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$; $\Delta_r H^0_{300} = -483\,kJ$

A) 111.5 kJ B) -109.01 kJ

C) -111.5 kJ D) -114 kJ

32.

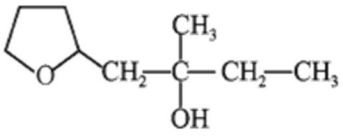


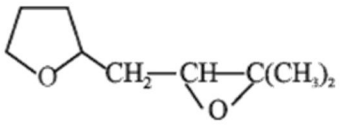
(X)

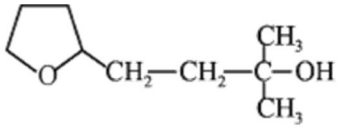
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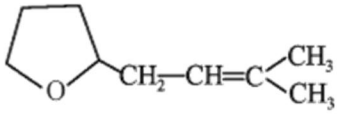
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Product of the reaction is

A) 

B) 

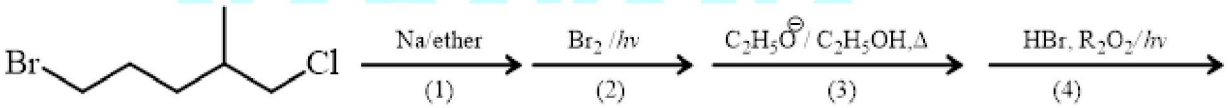
C) 

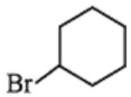
D) 

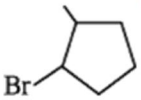
33. At constant temperature and pressure, the percentage of butane by volume in a mixture of butane, propane and propyne is 40%. What will be the volume of $CO_{2(g)}$ (in ml) under same pressure and temperature produced, when $150\,ml$ of gaseous mixture is burnt in excess of O_2?

A) 270 ml B) 150 ml C) 510 ml D) 240 ml

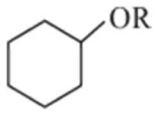
34. The major final product in the following reactions sequence is:



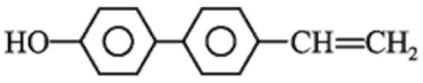
A) 

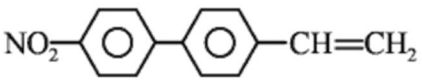
B) 

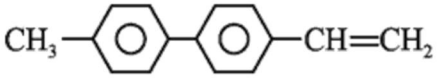
C) 

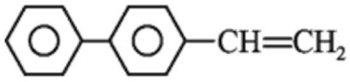
D) 

35. Slowest rate of electrophilic addition will take place in

A) 

B) 

C) 

D) 

36. Among the following, the species that is both paramagnetic and coloured is

A) $[MnO_4]^{2-}$ B) $[TiCl_6]^{2-}$ C) $[VO_4]^{3-}$ D) CrO_2Cl_2

APEX JEE-MAIN 2025

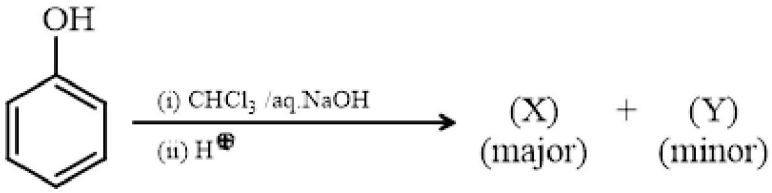
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37. Total number of possible isomers (including stereo) for $\left[Fe(gly)_2Cl_2\right]^{-2}$

$\left[gly = NH_2 - CH_2 - COO^-\right]$

A) 4 B) 6 C) 8 D) 2

38.



(X) or (Y) $\xrightarrow[\Delta]{Zn\ dust}(Z)+ZnO$. 'Z' cannot be prepared by

A) $C_6H_5COCl \xrightarrow[Pd-BaSO_4]{H_2}$ B) $C_6H_5CN \xrightarrow[(ii)\ H_3O^+]{(i)SnCl_4+HCl}$

C) $C_6H_5CH_3 \xrightarrow[(ii)H_3O^{\oplus}]{(i)CrO_2Cl_2/CS_2}$ D) $C_6H_5CH_3 \xrightarrow[(ii)Aq.KOH]{(i)NBS(2eq),h\upsilon}$

39. Consider the following compounds and select the **correct** statement from the following:

$NH_3, PH_3, SO_2, BF_3, NO_2^-, ClO_3^-, CO$

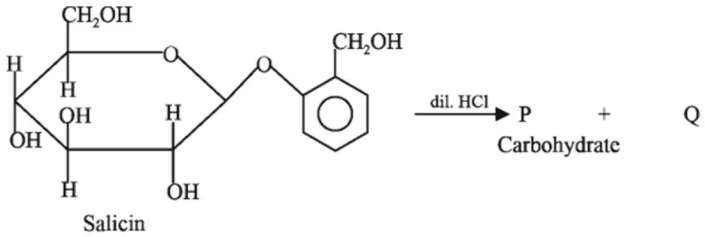
A) Three molecules out of given compounds involves sp^2 hybridization for central atom

B) Two molecules out of given compounds involves sp^3d hybridization for central atom

C) Four molecules out of given compounds involves both $(p\pi - d\pi)$ and $(p\pi - p\pi)$ bonding

D) Four molecules out of given compounds are hypervalent compounds

40. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain. Observe the following reaction of salicin



Salicin

The correct statement(s) is (are) :

I. P is D- glucose

II. Q is 2-hydroxybenzylalcohol

APEX JEE-MAIN 2025

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III. P is L-glucose

IV. The above reaction occurs through a carbocation

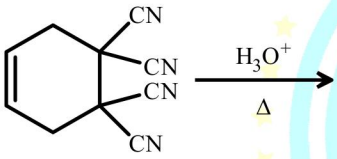
A) I, II, III only B) I, II, IV only C) I, IV only D) I, II only

41. Identify correct order of increasing ionic radii.

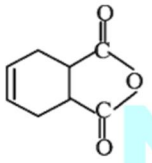
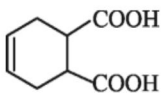
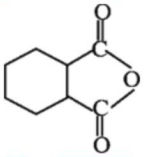
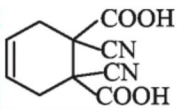
A) $Mg^{2+}, Na^{+}, F^{-}, O^{2-}, N^{3-}$ B) $N^{3-}, O^{2-}, F^{-}, Na^{+}, Mg^{2+}$

C) $F^{-}, Na^{+}, O^{2-}, Mg^{2+}, N^{3-}$ D) $Na^{+}, F^{-}, Mg^{2+}, O^{2-}, N^{3-}$

42.



A; Product (A) of the reaction is

A)  B)  C)  D) 

43. The incorrect order of the following

A) $B > Tl > Ga > Al > In$, (IP_1 order)

B) $C > Si > Ge > Sn > Pb$ (EN order)

C) $S > Se > Te > O$ (Negative electron gain enthalpy order)

D) $Cl_2 > Br_2 > F_2 > I_2$ (Bond energy order)

44. Identify the f-subshells are half and completely filled, respectively in lanthanide ions.

[Given : Atomic no. Eu, 63 ; Sm, 62; Tm, 69 ; Tb, 65 ; Yb, 70 ; Dy, 66]

A) Eu^{2+} and Tm^{2+} B) Sm^{2+} and Tm^{3+} C) Tb^{4+} and Yb^{2+} D) Dy^{3+} and Yb^{3+}

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45. Match the complex ions listed in column –I with the characteristics listed in column – II using the codes gives below.

	Column I		Column II
(a)	$\left[Ni\left(NH_3\right)_6\right]^{2+}$	(p)	d^2sp^3 and diamagnetic
(b)	$\left[CoF_6\right]^{3-}$	(q)	sp^3d^2 and diamagnetic
(c)	$\left[Co\left(NH_3\right)_6\right]^{3+}$	(r)	sp^3d^2 and two unpaired electrons
(d)	$\left[Zn\left(NH_3\right)_6\right]^{2+}$	(s)	sp^3d^2 and four unpaired electrons.

A) $a \to s; b \to r; c \to q; d \to p$ B) $a \to r; b \to s; c \to p; d \to q$

C) $a \to r; b \to s; c \to q; d \to p$ D) $a \to p; b \to q; c \to r; d \to s$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. How many condensation products including stereoisomers are possible when ethanal and phenyl ethanal (mixture) is treated with dil. NaOH/Δ ?

47. X = Sum of antibonding π electrons (π^* electrons) in species O_2, O_2^- and O_2^{2-}

Y = The number of species, among the following species, where central atoms have two lone pairs of electrons

XeF_4	XeF_5^-	F_2SeO_2
XeF_3^+	$XeOF_4$	$ClOF_3$
ICl_4^-	SCl_2	OSF_4

Find the value of X + Y.

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48. Given cell is $H_{2(g)}\left(P=1.0atm\right)\left|H^{+}_{(aq)}\left(pH=X\right)\right|\left|Calomel\left(1N\right)\right|$. At 25^0C, the standard electrode potential is 0.28V and the cell potential is 0.516 V. What is the value of X?

(Given: $\frac{2.303RT}{F}=0.059$)

49. If Phosphorus acid, Tetrathionic acid and Pyrophosphoric acid have maximum number of acidic hydrogens per molecule respectively are x, y and z, then find the value of x + y + z?

50. In a linear tetrapeptide (Constituted with different amino acids), then, number of amino acids $=x$, number of peptide bonds $=y$ then $x+y=?$

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. $\int e^x\left(\frac{1+10x^9-x^{20}}{\left(1-x^{10}\right)\sqrt{1-x^{20}}}\right)dx$ is equal to: (where c is an integral constant)

A) $e^x\left(\frac{1-x^{10}}{1+x^{10}}\right)+c$ B) $e^x\left(\frac{1+x^{10}}{1-x^{10}}\right)+c$ C) $e^x\left(\sqrt{\frac{1+x^{10}}{1-x^{10}}}\right)+c$ D) $e^x\left(\sqrt{\frac{1-x^{10}}{1+x^{10}}}\right)+c$

52. The value of $\lim_{n\to\infty} n^{n^{-n}}\left(\left(n+1\right)\left(n+\frac{1}{3}\right)\left(n+\frac{1}{3^2}\right)....\left(n+\frac{1}{3^{n-1}}\right)\right)^n$ is e^k then k is

A) 1.5 B) 2.5 C) 1 D) 0

53. If $\cos\alpha,\sin\beta,\sin\alpha$ are in increasing G.P., then roots of $x^2-2\cot\beta.x+1=0$ are: (where $\alpha,\beta\in R$)

A) Real and distinct B) both positive C) Real and equal D) Imaginary

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54. Let $\Delta=\begin{vmatrix} -bc & b^2+bc & c^2+bc \\ a^2+ac & -ac & c^2+ac \\ a^2+ab & b^2+ab & -ab \end{vmatrix}$ and the equation $px^3+qx^2+rx+s=0$ has roots a, b, c where $a,b,c\in R^+$. If $\Delta=27$ and $a^2+b^2+c^2=2$, then $\frac{\sqrt{2}p}{q}+1$ is

A) 1 B) 0 C) 1/2 D) – 1/2

55. In a test an examinee either guesses, copies or knows the answer to multiple choice question with five choices. The probability that he makes a guess is $\frac{1}{4}$ and probability that he copies the answer is $\frac{1}{8}$. The probability that his answer is correct given that he copies is $\frac{1}{10}$. Find the probability that he knows the answer to the question given that he correctly answered it:

A) 1 B) $\frac{10}{11}$ C) $\frac{1}{2}$ D) $\frac{12}{19}$

56. The value of $\int_{-20\pi}^{20\pi}[\sin x+\cos x]dx$ is: (where [.] denotes greatest integer function)

A) 10π B) -20π C) 20π D) -10π

57. The area of the parallelogram formed by the lines $4y-3x=1,\ 4y-3x-3=0,\ 3y-4x+1=0,\ 3y-4x+2=0$ is

A)3/8 B) 2/7 C) 1/6 D) 5/6

58. If a, b, c are real, then $f(x)=\begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix}$ is decreasing in

A) $\left(-\frac{2}{3}\left(a^2+b^2+c^2\right),0\right)$ B) $\left(0,\frac{2}{3}\left(a^2+b^2+c^2\right)\right)$

C) $\left(0,\frac{a^2+b^2+c^2}{3}\right)$ D) No where

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59. The function $f:R\to\{x\in R:-1<x<1\}$ defined by $f(x)=\frac{x}{1+|x|}$ is

A) One – one and into
B) One – one and onto
C) many one and into
D) many – one and onto

60. If a conic passes through (1, 0) and satisfies differential equation $(1+y^2)dx-xydy=0$. Then the equation of circle which is touching this conic at $(\sqrt{2},1)$ and passing through one of its foci is:

A) $x^2+y^2-3\sqrt{2}x-y+4=0$
B) $x^2+y^2-\sqrt{2}x-y+1=0$
C) $x^2+y^2-\sqrt{2}x-3y+1=0$
D) $x^2+y^2-3\sqrt{2}x-3y+4=0$

61. Area bounded by the graphs of $y=x^2-1$, $y=0$, $x=0$ and $x=2$

A) 14/3
B) 3
C) 2
D) 2/3

62. An ellipse passing through the point $(2\sqrt{13},4)$ has its foci at (-4, 1) and (4, 1). Then its eccentricity is

A) $\frac{1}{2}$
B) $\frac{1}{\sqrt{2}}$
C) $\frac{1}{4}$
D) $\frac{1}{2\sqrt{2}}$

63. A (1, 0) and B (0, 1) are two fixed points on the circle $x^2+y^2=1$. C is a variable point on this circle. As C moves, the locus of the ortho centre of the triangle ABC is

A) $x^2+y^2-2x-2y+1=0$
B) $x^2+y^2-x-y=0$
C) $x^2+y^2=4$
D) $x^2+y^2+2x-2y+1=0$

64. In binomial probability distribution B(n, p), mean is 3 and standard deviation is $\frac{3}{2}$. Then the value of p is

A) $\frac{1}{4}$
B) $\frac{3}{4}$
C) $\frac{1}{2}$
D) $\frac{1}{3}$

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65. If the Second term of the expansion $\left[a^{\frac{1}{13}}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$ then the value of $\frac{{}^nC_3}{{}^nC_2}$ is

A) 4 B) 3 C) 12 D) 6

66. If p^{th}, q^{th}, r^{th}, terms of G.P. are the positive number a, b, c respectively, then angle between the vectors $\log a^3\hat{i}+\log b^3\hat{j}+\log c^3\hat{k}$ and $(q-r)\hat{i}+(r-p)\hat{j}+(p-q)\hat{k}$ is

A) $\frac{\pi}{2}$ B) $\frac{\pi}{3}$ C) 0 D) $\sin^{-1}\left(\frac{1}{\sqrt{p^2+q^2+r^2}}\right)$

67. Let $\vec{a}=\hat{i}+\lambda\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+\hat{k}$. If $\left|\vec{a}+\vec{b}\right|=\left|\vec{a}\right|+\left|\vec{b}\right|$ and $f(x)=\int_0^{\sqrt{x}}\sin\left(t^2\right)dt$, then the value of $f'(\lambda)=$

A) $\frac{1}{4}\sin(4)$ B) $\frac{1}{2\sqrt{2}}\sin(2)$ C) $\frac{1}{\sqrt{2}}\sin\left(\frac{1}{2}\right)$ D) $\frac{1}{2}\sin(1)$

68. Let $f(x)=x+\sin x$. Suppose g denotes the inverse function of f. The value of $g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)$ is equal to:

A) $\sqrt{2}-1$ B) $\frac{\sqrt{2}+1}{\sqrt{2}}$ C) $2-\sqrt{2}$ D) $\sqrt{2}+1$

69. Let $P=\{x:x\in R,|x|<2\}$
$Q=\{x:x\in R,|x-1|\geq 2\}$

$P\cup Q=$ R- S then set S is: (where R is the set of real numbers)

A) [-2, 2] B) [2, 3] C) [2, 3) D) (-1, 2]

70. Equation of the line of shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is

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A) $\frac{x-\frac{5}{3}}{2}=\frac{y-3}{1}=\frac{z-\frac{13}{3}}{-1}$

B) $\frac{x-\frac{5}{3}}{1}=\frac{y-3}{2}=\frac{z-\frac{13}{3}}{-1}$

C) $\frac{x-\frac{5}{3}}{2}=\frac{y-3}{-1}=\frac{z-\frac{13}{3}}{1}$

D) $\frac{x-\frac{5}{3}}{1}=\frac{y-3}{-2}=\frac{z-\frac{13}{3}}{1}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. The number of values of x satisfying $2^{\log_5 16.\log_4 x+\log_{\sqrt{2}} 5}+5^x+x^{\left(\log_5 4\right)+5}+x^5=0$

72. The equation $z^2-(3+i)z+(m+2i)=0\ \ (m\in R)$ has exactly one real and one non real complex root, then product of real root and imaginary part of non-real complex root is:

73. If $e^{x+y}=y^2$ then $\frac{d^2y}{dx^2}$ at $(-1,1)$ is equal to

74. Let a and b respectively be the points of local maximum and local minimum of the function $f(x)=2x^3-3x^2-12x$. If A is the total area of the region bounded by $y=f(x)$, the x – axis and the lines x = a and x = b, then 4A is equal to

75. If 'a' denotes the number of permutation of 'n' different things taken all at a time, 'b' the number of permutation of $(n-2)$ different things taken 10 at a time and 'c' be the number of permutations of $(n-12)$ different things taken all at a time such that $a=182bc$, then value of 'n' is:

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